

Jiayi Wang (James)

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

August 2025 - present

M.S. in Music Technology; GPA 4/4; Music Informatics Lab

Occidental College, Los Angeles, CA.

August 2020 - May 2025

B.A. in Music, Emphasis in Music Production; B.A. in Computer Science; Minor in Mathematics

GPA 3.88/4; **Magna Cum Laude**; Dean's list: All semesters to date;

Kurtich Music Grant: four semester of scholarship from 2023 to 2025

Publications, Presentations & Academic Service

Jiayi Wang and Alexander Lerch. "Full-Page Optical Music Recognition of Jazz Lead Sheets." Manuscript under review for the 27th International Society for Music Information Retrieval Conference, ISMIR 2026.

Stephen Hudson, Jiayi Wang. "Harmonic Divorce or Not? Subjectively Hearing Pedals as Multi-Layered Chords." *Music Theory Online*, Vol. 34, Issue 4 (Accepted; scheduled for December 2025 publication).

Jiayi Wang. (2023, August). "Cinematic Pedal in Rock and Pop Music." Occidental College Summer Research Conference, Los Angeles, CA

Jiayi Wang (2022, August). "Re-evaluating Use Cases in VIVO." Andrew W. Mellon Summer Faculty Institute Conference 2022, Los Angeles, CA

RESEARCH EXPERIENCE

Research Seminar, Optical Music Recognition for Jazz Lead Sheet: [Github](#)

August 2025 - present

- Proposed **the first end-to-end system for full-page handwritten jazz lead sheet recognition**, leveraging a Sheet Music Transformer to jointly transcribe melody and chord symbols from images.
- Designed a **9-stage curriculum learning strategy** to progressively scale from single-staff inputs to full-page transcription using ~1.8k samples.
- Developed a detect-and-concatenate baseline using YOLO-based staff detection followed by staff-level recognition and reconstruction, and introduced chord-focused evaluation metrics.
- Achieved **substantial improvements** on a 32-page handwritten test set, reducing chord symbol error rate from 47.8% to 30.7% ($p < 0.001$), significantly outperforming the baseline.

Independent Research, Real-Time Human-Robot Improvisation System: [Github](#)

January 2026 - Present

- Developed a **real-time human-robot improvisation system** based on the Anticipatory Music Transformer (AMT), extending it from offline MIDI continuation to interactive co-performance with a marimba-playing robot.
- Fine-tuned AMT on the PiJAMa jazz improvisation MIDI dataset to better align generated output with jazz improvisational style and human musical interaction.
- Designed a real-time pipeline where human MIDI input is streamed via OSC to a cloud GPU for inference and Viterbi decoding, then transmitted back for robotic actuation.

Audio Content Analysis Project, Audio to Audio Patterns for Prediction: [Github](#), [PDF](#) September - December 2025

- Implemented an **end-to-end audio-to-audio music continuation system** for the MIREX *Patterns for Prediction* task, operating directly on raw audio without symbolic MIDI representations
- Designed a full preprocessing pipeline including **tempo normalization, spectrogram slicing, feature extraction, and clustering** to enable sequence modeling on musical audio
- Compared multiple modeling strategies, including **Gaussian HMMs, categorical HMMs, and a custom variable-order HMM**, to capture short- and long-range musical structure
- **Proposed a variable-order Hidden Markov Model over clustered musical features (chroma, pitch, RMS)**, enabling longer contextual conditioning and improved structural coherence in generated continuations
- Evaluated generated continuations using **MIREX Cardinality and Pitch Scores**, achieving performance comparable to symbolic Markov baselines while highlighting limitations of symbolic metrics for audio-based systems

Audio Content Analysis Project, Deep Salient Detection for F0 Estimation: [Github](#), [PDF](#) September - December 2025

- Replicated and evaluated a **deep learning–based pitch tracking system** for frame-wise fundamental frequency (F0) estimation, based on Bittner et al. (ISMIR 2017)
- Implemented an **HCQT-based audio representation** to explicitly encode harmonic structure, enabling accurate salience learning across multiple harmonic channels
- Designed and trained a **fully convolutional neural network (SimpleNet)** for dense time–frequency salience prediction, preserving pitch localization without pooling operations
- Conducted a comparative study between **rule-based autocorrelation, CNN-based salience detection, and transfer learning with pretrained ResNet18**, analyzing architectural inductive biases for audio tasks
- Evaluated models using standard **mir_eval melody metrics** (VR, VFA, RPA, RCA, OA), achieving the highest Overall Accuracy (81.6%) with the domain-specific CNN and demonstrating limitations of vision-pretrained models for spectral pitch estimation

Undergraduate Thesis, Music Dept. Senior Comprehensive Project: [Github](#), [PDF](#)

June - December 2024

Tracing the Evolution of Rock and Popular Music Harmony: A Computational Study Using Shannon Entropy

- **Designed and executed** a computational musicology study analyzing 890 rock songs (1950s–2020s) from the McGill Billboard dataset.
- **Implemented Shannon entropy** as harmonic complexity metric, demonstrating the viability of information theory as a quantitative framework for genre evolution research.
- **Discovered a measurable rise and fall in harmonic complexity**, peaking in late-70s progressive rock before declining in the 1980s and diversifying through the 1990s–2000s
- **Presented** results at Occidental College’s Music Department Senior Project Conference.

Undergraduate Thesis, Computer Science Dept. Senior Comprehensive Project: [Github](#), [PDF](#) June - December 2024

Reference-free metric for evaluating audio stem separation quality

- **Developed a novel reference-free evaluation method** for audio stem separation systems, eliminating reliance on original ground-truth stems.
- **Designed and implemented two new metrics** — Frequency Isolation Score (FIS) for spectral leakage detection and Dynamic Stability Score (DSS) for temporal consistency analysis.
- **Evaluated on the MUSDB18-HQ dataset** (150 professionally mixed songs) and demonstrated strong alignment over traditional SDR/SIR/SAR metrics.
- **Presented** results at Occidental College’s Computer Science Senior Project Conference.

Research Assistant, Video Interactive VST Orchestra (VIVO) Implementation and Enhancement

VIVO is an interactive music system developed by Fabio Paolizzo in Max/MSP that takes user audio and video inputs, maps the visual data to VST controls to modify the audio in real-time, and outputs the enhanced audio.

Principal Investigator: Dr. Fabio Paolizzo

Funding: European Commission Horizon Grant 2020; Mellon Foundation Grant

AI Orchestra – A Fusion of Electronic and Classical Music

August 2022 – December 2024

- Designed and ran tests on VIVO to digitize a conductor's hand movements to allow touchless control over electronic music parameters, including dynamics, filter cutoff frequency, filter Q value, reverb amount, compression threshold, distortion gain, and flanger/chorus frequency and amount.

Real-Time VST Parameter Prediction

August 2021 – December 2024

- Developed a component in MAX/MSP using Hierarchy Hidden Markov Chain Model that allows VIVO to simulate VST parameter changes according to previous video/audio input on a real-time basis, eliminating the need for continuous data input.

AI Sound Recognition Integration

August 2021 – December 2024

- Gathered audio data from BOOM one package music library (48,000+ files containing 125,000+ sound effects).
- Wrote a JavaScript program that transformed each volume peak and its timing into a density score for 70% of the library data for training.
- Trained a Hierarchy Hidden Markov Chain Model to recognize different sounds based on audio input from VIVO; will use the rest 30% of the library data to test the model's accuracy.

VIVO Use Case Study

February - April 2022

- Tested the VIVO system; identified bugs and areas for improvement in workflow.
- Evaluated five use cases including (1) connect physical movements to software instruments, (2) map physical gestures on physical interfaces to software instruments, (3) reflect hand movements on physical musical instruments on software instruments, (4) record interaction on virtual interface within software instruments, and (5) instruct changes in software instruments by programming languages.
- Presented the use case study at the Andrew W. Mellon Summer Faculty Institute Conference in 2023.

Co-investigator, Occidental Summer Research Program, Los Angeles, CA: [PDF](#)

June - August 2023

Cinematic Pedal in Rock and Pop Music: How their use in contemporary music deviates from traditional textbook definitions and how they affect different music genres; Advisor: Dr. Stephen Hudson

- Transcribed 20 distinct pedal examples across rock and pop music to illustrate real-world applications.
- Categorized four distinct uses of pedal tones and analyzed their effectiveness and impact across rock, pop, and classical musical genres.
- Presented the research at the Occidental College Summer Research Conference
- Submitted a journal article to *Music Theory Online* with Dr. Hudson that explores the diverse perceptions of sound in pedal tones (scheduled to publish in December 2025)

Machine Learning Research: [Github](#), [PDF](#)

September - December 2023

Music Genre Recognition: Exploring what model and how many audio parameters can most accurately classify genres

- Extracted a set of 6,000 parameters using the openSMILE package from the GTZAN dataset (1000 audio tracks each 30 seconds long, spanning 10 genres with each genre representing 100 tracks)
- Trained and tested four machine learning models: Support Vector Machine (SVM), Random Forest, Stack Ensemble, and Multilayer Perceptron (MLP)
- Evaluated the training results based on classification accuracy and confusion matrix; concluded that the MLP algorithm performed optimally, achieving an 89% accuracy with 2,000 to 3,000 audio parameters.

Research Assistant, Occidental College Computer Science Dept., Los Angeles, CA *November 2021 – May 2022*
Category Theory: exploring the intersection between category theory, operads, and neural network dimensions; Advisor:
Principal Investigator: Dr. Paul David

- Independently studied the principles of category theory and operads; learned to use the MNIST dataset.
- Programmed Convolution Neural Network; no significant correlation detected between the dimensions of a neural network and category theory

MUSIC SOFTWARE DEVELOPMENT

Software developer, Gesture Ensemble Music System (GEMS): [Github](#), [PDF](#) *February 2024 - Present*

- Designed and implemented **GEMS**, a gesture-controlled human-machine improvisation system enabling a solo performer to interact with a responsive virtual ensemble in **real time** using Max/MSP.
- Built **genre-adaptive generative engines** (chords, bass, drums) driven by symbolic chord progressions, providing harmonically grounded accompaniment for improvisation practice.
- Implemented real-time **gestural control** of music generation using MediaPipe-based pose tracking, streaming performer movement data via **UDP** to influence generative behavior and ensemble interaction.

Software Developer, VST Plug-in Development project *June - August 2023*

- Surveyed the faculty and students at the Occidental College Music Department to form the idea of developing a comprehensive vocal chain processor VST, featuring EQ, compressor, gate, and de-esser, to replace using multiple plug-ins that would require fixing individual vocal stems with each plug-in.
- Followed a JUCE framework tutorial to develop a 3-band EQ plug-in, applying classroom learning such as C++ and digital signal processing.
- Designed an original user interface for this EQ plug-in, drawing inspiration from industry leaders (Fabfilter, Valhalla DSP).
- Implemented vocal chain processor and integrated this VST into 5 people's music production projects; saved 20 minutes on average for each project.

Audio Interface Designer, Huaxin Musical Instrument, Los Angeles, CA *June - August 2024*

- Designed a user interface resembling a soundboard in Blender for the live streaming and recording functions of an electronic piano and drum kit.
- Drafted product specifications featuring three microphone preamps with one XLR input and two-line inputs, software sound drive that enables VST processing, multiple supported sample rate and bit-depth, and USB-C connectivity; adjusted detailed parameters such as gain range, impedance, and maximum input for each input and output to balance high performance with affordability.
- Worked with engineers to clarify design intentions, ensuring the final product reflected the specifications.

TEACHING EXPERIENCE

Music Department Peer Tutor, Occidental College, Los Angeles, CA *September 2024 - May 2025*

- Hold office hours for 40 students regarding academic questions in music theory/production students
- Address advising and registration questions with new music majors

MUSIC RECORDING AND PRODUCTION

- [Senior Comprehensive Project Album](#): Player; Recording, Mixing, Mastering Engineer; 11 songs (2024)
- [Occidental College Chamber Jazz Album 2024](#): Player; Recording, Mixing, Mastering Engineer; 11 songs (2024)
- [Occidental College Chamber Jazz Album 2023](#): Player; Recording, Mixing, Mastering Engineer; 10 songs (2023)
- [Occidental Music YouTube Channel](#): edited the audio and visual component of 15 recorded performances and published them on the Music Department's official YouTube channel (2022)
- “[Berret Yuffee's EP](#)” (three songs composed by an Occidental student): recorded vocal and guitar; mixed and mastered in Pro Tools (2024)

MUSIC PERFORMANCE

- Bowers Museum member's opening night reception performance: tenor saxophone and electronics (2024)
- [Occidental College Jazz Ensemble](#): Tenor saxophone & tech support; 10 total performances; two per semester (since 2023)
- [Occidental College Chamber Jazz](#): Alto saxophone; 20 total performances; two to three per semester (since 2021)
- [Occidental College Symphony Orchestra](#): Tuba; six total performances; three per school year (2021 – 2023)
- Chouby (Occidental student-led rock band): Guitar and drums; 4 total performance, one per school year (2021 – 2023)

MUSIC STUDIO MANAGEMENT

Production Manager Assistant, Occidental College Music Department, Los Angeles, CA *January 2022 – May 2025*

- Assist with event promotion, logistics, and social media management for on-campus live musical performances (15-20 events per semester).
- Live Stream and Record events on [YouTube](#), including Set up/Tier down live stream equipment and monitor the soundboard, OBS streaming system, and other audio equipment.
- Manage inventory (cameras, microphones, speakers, soundboards, cables); create a labeling system for equipment, and maintain a digitized archive of past performance materials.

Studio Assistant, Occidental College Choi Studio Assistant, Los Angeles, CA *August 2023 – May 2025*

- Manage Weekly inventory (worth \$60K+), including checking in and out equipment, monitoring equipment status after returns; maintain studio and storage space, Install and ensure software is ready for use.
- Supervise and Provide technical support for students and enforce the studio's leasing policies (2-3 three hour session per week)
- Designed and managed EMS(Electronic Music Studio) at Occidental College including installing audio equipment, manage software update, and coordinate work orders cross departments (Music, IT, and Maintenance)

Recording Intern, Kingsize Soundlabs, Los Angeles, CA *February – June 2023*

- Shadowed professional engineers in recording sessions, including collaborations with Lyla Forde, Will Brahm, Walter Trout, Aidan Carroll, Julius Rodriguez, Drown, and Dudley.
- Assisted in studio operations, including assembling and disassembling recording equipment, daily studio cleanup and maintenance, managing inventories, maintaining the single VIP vocal studio, and coffee and lunch runs.

Music Production Intern, Jiali Music Studio, Shanghai, China *April – June 2021*

- Prepare client meetings to learn about clients' needs for OST, commercial, video game music

DATA & BUSINESS INTERNSHIPS

Data Science Intern, Leizhi Information Technology, Shanghai, China

July – August 2024

Clustering and Density Analysis of High-Risk Merchants in Shanghai Pudong District

- Acquired Shanghai Pudong District merchant data with multiple rectifications from the government's database (SQL).
- Cleaned data by removing unwanted fields, encoding Chinese characters, and manual searching for missing data.
- Ran K-means and KDE algorithm to get cluster and location density information.
- Summarized merchant characteristics by clusters; produced a heat map that displays the density of high-risk merchants in Pudong District for government policymaking.

Temporal Analysis and Real-Time Monitoring System Development for Merchant Complaints

- Extracted and cleaned data on merchant's unauthorized use of public spaces from Pudong Urban Management Complaints Platform.
- Grouped complaint data by hour, day, and week; applied z-score to display complaint hotspots over time (days over weeks and month, hours over days).
- Visualized the hotspots of complaints by day, week, and month that feed into the government's real-time monitoring system.

Consultant Intern, Ernst & Young, Shanghai, China

June – July 2021

- Compiled government policies for E&Y's annual report on China's education industry
- Researched 3D cell printing technology about its development and application for a due diligence report
- Researched Istanbul Airport's transfer service to advise Shanghai Pudong Airport on its initiative to boost travel industry.
- Gathered existing cases of comparable factories for a chemical company's development plan.

Lombarda China Fund Management, Shanghai, China

December 2020 – January 2021

Finance Intern

- Self-studied accounting fundamentals and learned to implement the DCF model.
- Reported on audit fees of the top 10 fund management companies in China; analyzed and summarized the fee structures to provide reference for Lombarda's pricing strategy.

SKILLS & INTERESTS

- Programming: Java, Python, C/C++, HTML/CSS, MAX/MSP, SQL, Data visualization, Blender, Latex, Mathematica
- Music production: DAWs (Logic Pro X, Ableton Live, Pro Tools), Izotope RX, Izotope ozone, Melodyne
- Instruments: Saxophone (10+years), Guitar, Piano, Tuba, and Drums
- Treasurer of the Occidental Chinese Students & Scholars Association: managing \$5000 each year; organized lunar year celebrations, bi-weekly movie nights, concerts, bonfire parties, new student social events
- Student Representative for SUNO AI at Occidental College
- Language: Chinese Mandarin (Native) and English (Fluent)
- Interests: e-sport (League of Legends North America top 20%), Oxy Intramural Basketball Team, Snowboarding (black diamond, freestyle park)